REINFORCEMENT LEARNING-BASED MATERIAL SCREENING

Using the Epsilon-Greedy Multi-Armed Bandit Algorithm

Project Area	Reinforcement Learning
Unit	Unit 5 – Real-World RL
Selected Topic	Bandit-Based Material Screening
Algorithm	Epsilon-Greedy Multi-Armed Bandit
Comparison	Random Selection Baseline

Prepared as an academic and educational simulation report

Student Name: _____

Course / Section: _____

Abstract

This project demonstrates how a multi-armed bandit method can support sequential candidate selection in a simplified material-screening setting. Eight candidate materials are represented by unknown reward distributions. An Epsilon-Greedy agent repeatedly selects a candidate, receives a noisy simulated performance reward, and updates its estimate of that candidate's average reward. Its cumulative reward is compared with a random-selection baseline over 30 runs of 500 tests each. The experiment illustrates exploration and exploitation; it does not use laboratory measurements or establish real material properties.

1. Introduction

Material screening can involve evaluating many candidate alloys or compounds to identify promising options. When each test is costly, a sequential decision strategy can use earlier observations to guide later selections. A multi-armed bandit is a simple reinforcement-learning framework for this setting: each candidate is an arm, selecting an arm is an action, and the observed performance is the reward.

2. Problem Formulation

Element	Definition in this project
Environment	A simulated screening process with 8 candidate materials.
State / information	The agent's estimated average reward and selection count for each material.
Action	Select one of the 8 candidate materials for the next test.
Reward	A noisy simulated performance observation for the selected material.
Objective	Maximize cumulative reward across sequential tests.

3. Methodology

3.1 Epsilon-Greedy action selection

At each test, the agent explores with probability $\varepsilon = 0.1$ by selecting a random material. With probability $1 - \varepsilon = 0.9$, it exploits by selecting the material with the highest current estimated reward. This balances trying less-known candidates with using the candidate that currently appears best.

3.2 Reward model and value update

Each candidate has a fixed, hidden expected reward in the simulation. When candidate a is selected, the observed reward is sampled from a normal distribution with that candidate's expected reward as its mean and standard deviation 1.0. The estimate is updated using the incremental sample-average rule:

$$\mathbf{Q(a) \leftarrow Q(a) + [R - Q(a)] / N(a)}$$

where Q(a) is the estimated value of candidate a, R is the newly observed reward, and N(a) is the number of times candidate a has been selected.

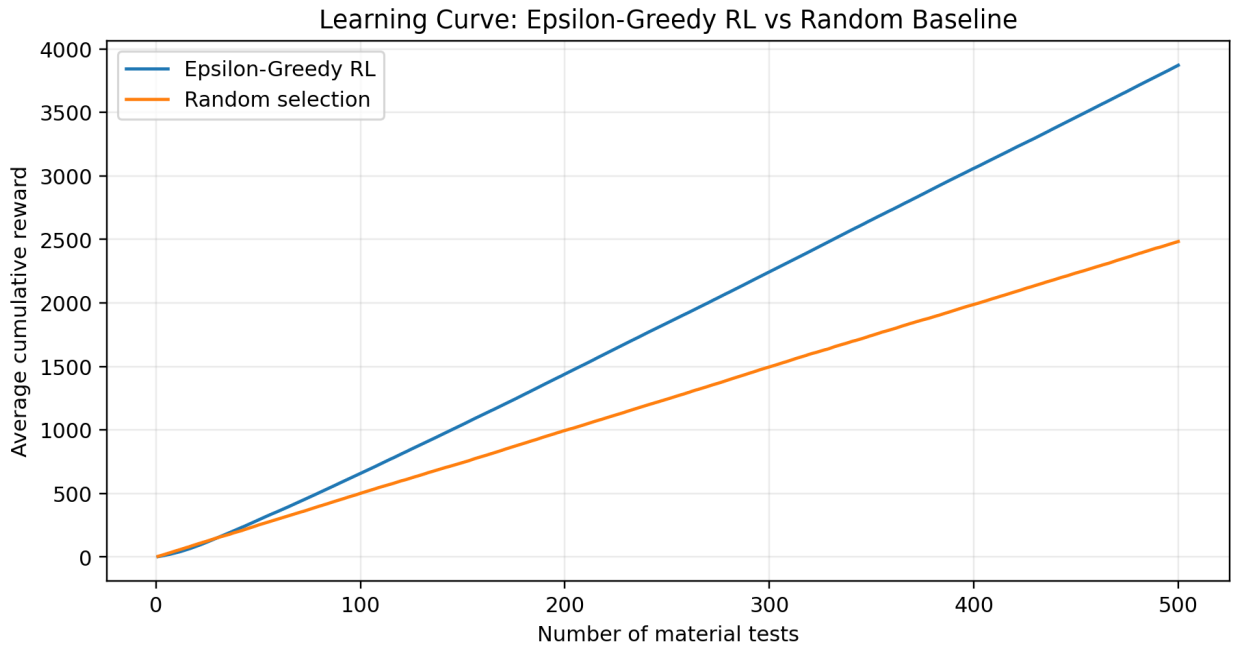
3.3 Experimental configuration

Parameter	Setting
Candidate materials	8
Tests per run	500

Parameter	Setting
Independent runs	30
Exploration probability (ε)	0.1
Reward noise standard deviation	1.0
Baseline	Uniform random material selection

4. Results and Discussion

The experiment records cumulative reward at each test and averages the learning curves across 30 independent runs. The plot below shows the comparison produced by the seeded simulation used for this report.



At the end of 500 tests, the mean cumulative reward was 3871.94 for Epsilon-Greedy and 2483.96 for random selection in this run. These are simulation-specific outcomes, not guaranteed values. The candidate with the highest configured expected reward is Material 8 (8.5), although the agent does not know these expected rewards in advance.

Interpretation

The Epsilon-Greedy method can improve its choices as it gathers observations. The random baseline continues selecting candidates without using the reward history. Consequently, the learned policy is expected to accumulate more reward when the reward differences are stable and sufficiently observable. The exact curve can vary with random sampling.

5. Conclusion

The project implements a simple reinforcement-learning strategy for sequential material screening. The Epsilon-Greedy bandit uses observed rewards to estimate candidate quality and balance exploration with exploitation. In the simulated experiment, it is compared with random selection using average cumulative reward. This provides a compact demonstration of how adaptive selection can be useful when testing multiple candidates.

6. Limitations

• The experiment uses synthetic reward distributions rather than experimental material data.
• The reward means are fixed and known only to the simulator; real material properties may be noisy, context-dependent, and expensive to measure.
• Only eight candidates and one scalar reward are considered.
• Epsilon is fixed at 0.1; the project does not optimize the exploration schedule.
• Results from this simplified simulation should not be interpreted as scientific claims about alloys or compounds.

7. Future Scope

Future work could evaluate the method on a public materials dataset, include material descriptors as context, compare Epsilon-Greedy with Upper Confidence Bound (UCB) or Thompson Sampling, and investigate Bayesian optimization. Any real-world use would require validated measurements and appropriate materials-science expertise.

8. Software and Reproducibility

The implementation uses Python, NumPy, and Matplotlib. Install dependencies with **pip install numpy matplotlib**, then run the project script with **python material_screening.py**. The script generates the learning-curve visualization. For exact reproduction, retain the same code, parameter settings, and random seed.

Appendix: Key Algorithm Steps

1. Initialize estimated rewards and selection counts to zero.
2. For each test, explore with probability ε; otherwise select the candidate with the largest estimate.
3. Sample a reward from the selected candidate's simulated reward distribution.
4. Increment that candidate's count and update its estimated average reward.
5. Record cumulative reward and repeat until all tests are complete.

End of Report